

Second Exercise Sheet for Week 5 - Abstract data types

Question 1: Type-checking with existential types

Give a derivation to show that each of the following is a valid typing judgement.

- (a) $\vdash \text{pack } \langle \mathbb{N} \times \mathbb{N}, \lambda p : \mathbb{N} \times \mathbb{N}. \langle \text{snd } p, \text{fst } p \rangle \rangle \text{ as } \exists \alpha. \alpha \rightarrow \alpha : \exists \alpha. \alpha \rightarrow \alpha$

$$\frac{\frac{\frac{\frac{}{p : \mathbb{N} \times \mathbb{N} \vdash p : \mathbb{N} \times \mathbb{N}}{\text{VAR}} \quad \frac{}{p : \mathbb{N} \times \mathbb{N} \vdash p : \mathbb{N} \times \mathbb{N}}{\text{VAR}}}{\frac{}{p : \mathbb{N} \times \mathbb{N} \vdash \text{snd } p : \mathbb{N}}{\text{Snd}}} \quad \frac{}{p : \mathbb{N} \times \mathbb{N} \vdash \text{fst } p : \mathbb{N}}{\text{Fst}}}{\frac{}{p : \mathbb{N} \times \mathbb{N} \vdash \langle \text{snd } p, \text{fst } p \rangle : \mathbb{N} \times \mathbb{N}}{\text{Pair}}} \quad \frac{}{\vdash \lambda p : \mathbb{N} \times \mathbb{N}. \langle \text{snd } p, \text{fst } p \rangle : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}}{\text{ABS}}}{\vdash \text{pack } \langle \mathbb{N} \times \mathbb{N}, \lambda p : \mathbb{N} \times \mathbb{N}. \langle \text{snd } p, \text{fst } p \rangle \rangle \text{ as } \exists \alpha. \alpha \rightarrow \alpha : \exists \alpha. \alpha \rightarrow \alpha} \text{Pack}$$

- (b) $\vdash \text{pack } \langle \mathbb{N}, \lambda p : \mathbb{N} \times \mathbb{N}. \langle \text{snd } p, \text{fst } p \rangle \rangle \text{ as } \exists \alpha. \alpha \times \alpha \rightarrow \alpha \times \alpha : \exists \alpha. \alpha \times \alpha \rightarrow \alpha \times \alpha$

This is exactly the same as the previous derivation except for the very last line. Recall that we typically have many choices for which interface (existential type) we export for a given concrete implementation.

$$\frac{\frac{\frac{\frac{}{p : \mathbb{N} \times \mathbb{N} \vdash p : \mathbb{N} \times \mathbb{N}}{\text{VAR}} \quad \frac{}{p : \mathbb{N} \times \mathbb{N} \vdash p : \mathbb{N} \times \mathbb{N}}{\text{VAR}}}{\frac{}{p : \mathbb{N} \times \mathbb{N} \vdash \text{snd } p : \mathbb{N}}{\text{Snd}}} \quad \frac{}{p : \mathbb{N} \times \mathbb{N} \vdash \text{fst } p : \mathbb{N}}{\text{Fst}}}{\frac{}{p : \mathbb{N} \times \mathbb{N} \vdash \langle \text{snd } p, \text{fst } p \rangle : \mathbb{N} \times \mathbb{N}}{\text{Pair}}} \quad \frac{}{\vdash \lambda p : \mathbb{N} \times \mathbb{N}. \langle \text{snd } p, \text{fst } p \rangle : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}}{\text{ABS}}}{\vdash \text{pack } \langle \mathbb{N}, \lambda p : \mathbb{N} \times \mathbb{N}. \langle \text{snd } p, \text{fst } p \rangle \rangle \text{ as } \exists \alpha. \alpha \times \alpha \rightarrow \alpha \times \alpha : \exists \alpha. \alpha \times \alpha \rightarrow \alpha \times \alpha} \text{Pack}$$

- (c) $\vdash \text{unpack } \alpha, x = P \text{ in } (\text{snd } x) (\text{fst } x) \text{ zero} : \mathbb{B}$
where $P = \text{pack } \langle \mathbb{N}, \langle \text{zero}, \lambda x : \mathbb{N}. \lambda y : \mathbb{N}. \text{true} \rangle \rangle \text{ as } \exists \alpha. \alpha \times (\alpha \rightarrow \mathbb{N} \rightarrow \mathbb{B})$.

First we'll type 'P'. Let Π_1 be the following derivation.

$$\frac{\frac{\frac{\frac{}{x : \mathbb{N}, y : \mathbb{N} \vdash \text{true} : \mathbb{B}}{\text{T}}}{\vdash \text{zero} : \mathbb{N}} \quad \frac{}{\vdash \lambda x : \mathbb{N}. \lambda y : \mathbb{N}. \text{true} : \mathbb{N} \rightarrow \mathbb{B}}{\text{Z}}}{\vdash \lambda x : \mathbb{N}. \lambda y : \mathbb{N}. \text{true} : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{B}} \quad \frac{}{\vdash \langle \text{zero}, \lambda x : \mathbb{N}. \lambda y : \mathbb{N}. \text{true} \rangle : \mathbb{N} \times (\mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{B})} \text{Pair}}{\vdash \text{pack } \langle \mathbb{N}, \langle \text{zero}, \lambda x : \mathbb{N}. \lambda y : \mathbb{N}. \text{true} \rangle \rangle \text{ as } \exists \alpha. \alpha \times (\alpha \rightarrow \mathbb{N} \rightarrow \mathbb{B}) : \exists \alpha. \alpha \times (\alpha \rightarrow \mathbb{N} \rightarrow \mathbb{B})} \text{Pack}$$

We will write $\Gamma = x : \alpha \times (\alpha \rightarrow \mathbb{N} \rightarrow \mathbb{B})$ to save space. Then our overall derivation is

$$\frac{\frac{\frac{\Gamma \vdash x : \alpha \times (\alpha \rightarrow \mathbb{N} \rightarrow \mathbb{B})}{\Gamma \vdash \text{snd } x : \alpha \rightarrow \mathbb{N} \rightarrow \mathbb{B}} \text{VAR} \quad \frac{\frac{\Gamma \vdash x : \alpha \times (\alpha \rightarrow \mathbb{N} \rightarrow \mathbb{B})}{\Gamma \vdash \text{fst } x : \alpha} \text{VAR}}{\Gamma \vdash (\text{snd } x) (\text{fst } x) : \mathbb{N} \rightarrow \mathbb{B}} \text{Fst}}{\Gamma \vdash (\text{snd } x) (\text{fst } x) \text{ zero} : \mathbb{B}} \text{Z} \quad \frac{\Pi_1 \quad \Gamma \vdash (\text{snd } x) (\text{fst } x) \text{ zero} : \mathbb{B}}{\vdash \text{unpack } \alpha, x = P \text{ in } (\text{snd } x) (\text{fst } x) \text{ zero} : \mathbb{B}} \text{APP} \quad \text{Unpack}$$

(d) $m : \exists \alpha. \alpha \times (\alpha \rightarrow \alpha) \times (\alpha \rightarrow \mathbb{B}) \vdash \text{unpack } \gamma, p = m \text{ in } (\text{snd } p) (\text{snd } (\text{fst } p) (\text{fst } (\text{fst } p))) : \mathbb{B}$

We'll write $\Gamma = m : \exists \alpha. \alpha \times (\alpha \rightarrow \alpha) \times (\alpha \rightarrow \mathbb{B}), p : \gamma \times (\gamma \rightarrow \gamma) \times (\gamma \rightarrow \mathbb{B})$ to save space.

$$\frac{\frac{\frac{\Pi_2}{\Gamma \vdash \text{snd } p : \gamma \rightarrow \mathbb{B}} \text{VAR} \quad \frac{\frac{\Pi_2}{\Gamma \vdash \text{fst } p : \gamma \times (\gamma \rightarrow \gamma)} \text{Fst} \quad \frac{\Pi_2}{\Gamma \vdash \text{fst } p : \gamma \times (\gamma \rightarrow \gamma)} \text{Fst}}{\Gamma \vdash \text{snd } (\text{fst } p) : \gamma \rightarrow \gamma} \text{Snd} \quad \frac{\Gamma \vdash \text{snd } (\text{fst } p) : \gamma \rightarrow \gamma \quad \frac{\Gamma \vdash \text{fst } (\text{fst } p) : \gamma}{\Gamma \vdash (\text{snd } (\text{fst } p)) (\text{fst } (\text{fst } p)) : \gamma} \text{APP}}{\Gamma \vdash (\text{snd } p) ((\text{snd } (\text{fst } p)) (\text{fst } (\text{fst } p))) : \mathbb{B}} \text{APP}}{\Gamma \vdash (\text{snd } p) ((\text{snd } (\text{fst } p)) (\text{fst } (\text{fst } p))) : \mathbb{B}} \text{Unpack} \quad \frac{\Pi_1 \quad \Gamma \vdash (\text{snd } p) ((\text{snd } (\text{fst } p)) (\text{fst } (\text{fst } p))) : \mathbb{B}}{m : \exists \alpha. \alpha \times (\alpha \rightarrow \alpha) \times (\alpha \rightarrow \mathbb{B}) \vdash \text{unpack } \gamma, p = m \text{ in } (\text{snd } p) ((\text{snd } (\text{fst } p)) (\text{fst } (\text{fst } p))) : \mathbb{B}} \text{Unpack}$$

where Π_1 is

$$\frac{}{m : \exists \alpha. \alpha \times (\alpha \rightarrow \alpha) \times (\alpha \rightarrow \mathbb{B}) \vdash m : \exists \alpha. \alpha \times (\alpha \rightarrow \alpha) \times (\alpha \rightarrow \mathbb{B})} \text{VAR}$$

and Π_2 is

$$\frac{}{\Gamma \vdash p : \gamma \times (\gamma \rightarrow \gamma) \times (\gamma \rightarrow \mathbb{B})} \text{VAR}$$

We have used the `Unpack` rule correctly given that existential types are intended to be read only up to ‘ α -equivalence’. This means that there is no need for the type variable γ which we use to unpack the package to match the bound type variable α used to denote the specification of m . We might as well say that

$$\exists \alpha. \alpha \times (\alpha \rightarrow \alpha) \rightarrow (\alpha \rightarrow \mathbb{B}) = \exists \gamma. \gamma \times (\gamma \rightarrow \gamma) \rightarrow (\gamma \rightarrow \mathbb{B}).$$

Question 2: ‘Type inference’ with existential types

For each of the following incomplete typing judgements in (Church-style) System F extended with (pairs and) existential types, fill in the holes where a missing type expression should be. (There may be multiple possible answers).

(a) $\vdash \text{pack } \langle \mathbb{B} \times \mathbb{N}, \langle \lambda x : \mathbb{B}. x, \lambda y : \mathbb{N}. \langle \text{iszero } y, y \rangle \rangle \rangle \text{ as } ?? : ??$

A typing derivation would need to begin

$$\frac{\vdots}{\vdash \langle \lambda x : \mathbb{B}. x, \lambda y : \mathbb{N}. \langle \text{iszero } y, y \rangle \rangle : T_2} \text{Pack} \quad \vdash \text{pack } \langle \mathbb{B} \times \mathbb{N}, \langle \lambda x : \mathbb{B}. x, \lambda y : \mathbb{N}. \langle \text{iszero } y, y \rangle \rangle \rangle \text{ as } T_0 : T_1$$

But for the pack rule we need $T_0 = T_1 = \exists\alpha.T$ for some type T , where $T_2 = T[\alpha \setminus (\mathbb{B} \times \mathbb{N})]$ is a substitution instance of T . Moreover, by straightforward type inference,

$$T_2 = (\mathbb{B} \rightarrow \mathbb{B}) \times (\mathbb{N} \rightarrow (\mathbb{B} \times \mathbb{N})).$$

So there are two possibilities for $T_0 = T_1$:

$$\begin{aligned} & \exists\alpha.(\mathbb{B} \rightarrow \mathbb{B}) \times (\mathbb{N} \rightarrow (\mathbb{B} \times \mathbb{N})) \\ & \exists\alpha.(\mathbb{B} \rightarrow \mathbb{B}) \times (\mathbb{N} \rightarrow \alpha) \end{aligned}$$

(b) $\vdash \text{pack } \langle \forall\alpha.\alpha \rightarrow \alpha, \lambda f : \forall\alpha.\alpha \rightarrow \alpha. \lambda x : \mathbb{N}.f\{\mathbb{N}\} x \rangle \text{ as } ?? : ??$

A typing derivation would need to begin

$$\frac{\vdots \quad \overline{\vdash \lambda f : \forall\alpha.\alpha \rightarrow \alpha. \lambda x : \mathbb{N}.f\{\mathbb{N}\} x : T[\alpha \setminus \forall\alpha.\alpha \rightarrow \alpha]}}{\vdash \text{pack } \langle \forall\alpha.\alpha \rightarrow \alpha, \lambda f : \forall\alpha.\alpha \rightarrow \alpha. \lambda x : \mathbb{N}.f\{\mathbb{N}\} x \rangle \text{ as } \exists\alpha.T : \exists\alpha.T} \text{Pack}$$

We continue the derivation backwards to find the possibilities for $T[\alpha \setminus \forall\alpha.\alpha \rightarrow \alpha]$.

$$\frac{\frac{\frac{f : \forall\alpha.\alpha \rightarrow \alpha, x : \mathbb{N} \vdash f : \forall\alpha.\alpha \rightarrow \alpha}{f : \forall\alpha.\alpha \rightarrow \alpha, x : \mathbb{N} \vdash f\{\mathbb{N}\} : \mathbb{N} \rightarrow \mathbb{N}} \text{TAPP} \quad \frac{f : \forall\alpha.\alpha \rightarrow \alpha, x : \mathbb{N} \vdash x : \mathbb{N}}{f : \forall\alpha.\alpha \rightarrow \alpha, x : \mathbb{N} \vdash f\{\mathbb{N}\} x : \mathbb{N}} \text{VAR}}{\frac{f : \forall\alpha.\alpha \rightarrow \alpha, x : \mathbb{N} \vdash f\{\mathbb{N}\} x : \mathbb{N}}{f : \forall\alpha.\alpha \rightarrow \alpha \vdash \lambda x : \mathbb{N}.f\{\mathbb{N}\} x : \mathbb{N} \rightarrow \mathbb{N}} \text{ABS}} \text{ABS}$$

So the two possibilities for $\exists\alpha.T$ are

$$\begin{aligned} & \exists\alpha.(\forall\beta.\beta \rightarrow \beta) \rightarrow \mathbb{N} \rightarrow \mathbb{N} \\ & \exists\alpha.\alpha \rightarrow \mathbb{N} \rightarrow \mathbb{N} \end{aligned}$$

(c) $f : \forall\alpha.\alpha \rightarrow \alpha \rightarrow \alpha \vdash \text{unpack } \theta, x = \text{pack } \langle \mathbb{N}, \langle \lambda n : \mathbb{N}.\text{pred } n, \lambda n : \mathbb{N}.\text{iszero } n \rangle \rangle \text{ as } ?? \text{ in } \text{snd } x (f\{\theta\} (\text{fst } x \text{ zero}) (\text{fst } x \text{ one})) : ??$

Let us first try to type the pack term. It doesn't use the variable f , so we'll ignore it for now.

$$\frac{\vdots \quad \overline{\vdash \langle \lambda n : \mathbb{N}.\text{pred } n, \lambda n : \mathbb{N}.\text{iszero } n \rangle : (\mathbb{N} \rightarrow \mathbb{N}) \times (\mathbb{N} \rightarrow \mathbb{B})}}{\vdash \text{pack } \langle \mathbb{N}, \langle \lambda n : \mathbb{N}.\text{pred } n, \lambda n : \mathbb{N}.\text{iszero } n \rangle \rangle \text{ as } ?? : ??} \text{Pack}$$

So the pack term will have type $\exists\alpha.(\beta_1 \rightarrow \beta_2) \times (\beta_3 \rightarrow \mathbb{B})$ where each β_i is either α or $\mathbb{N}$.

Now the unpack term is typed with the unpack rule. Let us abbreviate the pack term as P and the body of the unpack as M .

$$\frac{\overline{f : \forall\alpha.\alpha \rightarrow \alpha \rightarrow \alpha \vdash P : \exists\alpha.(\beta_1 \rightarrow \beta_2) \times (\beta_3 \rightarrow \mathbb{B})} \quad \overline{f : \forall\alpha.\alpha \rightarrow \alpha \rightarrow \alpha, x : ((\beta_1 \rightarrow \beta_2) \times (\beta_3 \rightarrow \mathbb{B}))[\alpha \setminus \theta] \vdash M : T}}{f : \forall\alpha.\alpha \rightarrow \alpha \rightarrow \alpha \vdash \text{unpack } \theta, x = P \text{ in } M : T} \text{Unpack}$$

Let us write $\gamma_i = \beta_i[\alpha \setminus \theta]$ and $\Gamma = f : \forall \alpha. \alpha \rightarrow \alpha \rightarrow \alpha, x : (\gamma_1 \rightarrow \gamma_2) \times (\gamma_3 \rightarrow \mathbb{B})$. We continue by trying to type $M : T$.

$$\frac{\frac{\frac{\Gamma \vdash x : (\gamma_1 \rightarrow \gamma_2) \times (\gamma_3 \rightarrow \mathbb{B})}{\Gamma \vdash \text{snd } x : \gamma_3 \rightarrow \mathbb{B}} \text{VAR}}{\Gamma \vdash \text{snd } x : \gamma_3 \rightarrow \mathbb{B}} \text{Snd} \quad \Pi_1}{\Gamma \vdash \text{snd } x (f\{\theta\}) (\text{fst } x \text{ zero}) (\text{fst } x \text{ one}) : \mathbb{B}} \text{APP} \boxed{3}$$

where Π_1 is

$$\frac{\frac{\frac{\Gamma \vdash f : \forall \alpha. \alpha \rightarrow \alpha \rightarrow \alpha}{\Gamma \vdash f\{\theta\} : \theta \rightarrow \theta \rightarrow \theta} \text{TAPP} \quad \frac{\frac{\frac{\Gamma \vdash x : (\gamma_1 \rightarrow \gamma_2) \times (\gamma_3 \rightarrow \mathbb{B})}{\Gamma \vdash \text{fst } x : \gamma_1 \rightarrow \gamma_2} \text{Fst} \quad \frac{\Gamma \vdash \text{zero}}{\Gamma \vdash \text{zero} : \gamma_2} \text{Z}}{\Gamma \vdash \text{fst } x \text{ zero} : \gamma_2} \text{APP} \boxed{2} \quad \frac{\Gamma \vdash \text{fst } x \text{ one} : \gamma_2}{\Gamma \vdash \text{fst } x \text{ one} : \gamma_2} \text{APP} \boxed{1}}{\Gamma \vdash f\{\theta\} (\text{fst } x \text{ zero}) (\text{fst } x \text{ one}) : \theta} \text{APP}$$

where we have omitted the branch of the tree on the right because it is so similar the middle branch. For this derivation to be valid, we need the following constraints to be satisfied (duplicates omitted).

$$\begin{array}{l} \boxed{1} : \gamma_1 = \mathbb{N} \\ \boxed{2} : \gamma_2 = \theta \\ \boxed{3} : \gamma_3 = \theta \end{array}$$

This completely determines the missing type expressions, giving a (unique) final answer of

$$\begin{array}{l} f : \forall \alpha. \alpha \rightarrow \alpha \rightarrow \alpha \vdash \\ \text{unpack } \theta, x = \text{pack } \langle \mathbb{N}, \langle \lambda n : \mathbb{N}. \text{pred } n, \lambda n : \mathbb{N}. \text{iszero } n \rangle \rangle \text{ as } \exists \alpha. (\mathbb{N} \rightarrow \alpha) \times (\alpha \rightarrow \mathbb{B}) \text{ in} \\ \text{snd } x (f\{\theta\}) (\text{fst } x \text{ zero}) (\text{fst } x \text{ one}) : \mathbb{B} \end{array}$$

(d) $O : ?? \vdash \text{unpack } \alpha, x = O \text{ in pack } \langle \alpha, \langle \text{if fst } x \text{ then false else true, snd } x \rangle \rangle \text{ as } ?? : ??$

We can see that the type of O must be $\exists \alpha. T_1$ for some T_1 . The final step in the whole type derivation must be

$$\frac{\frac{\vdots}{O : \exists \alpha. T_1, x : T_1 \vdash P : \exists \alpha. T_2}}{O : \exists \alpha. T_1 \vdash \text{unpack } \alpha, x = O \text{ in } P : \exists \alpha. T_2} \text{Unpack}$$

where P abbreviates the pack subterm. We will now attempt to type the term inside the pack, for which we can ignore the unused O variable.

$$\frac{\frac{\vdots}{x : T_1 \vdash \text{if fst } x \text{ then false else true} : \mathbb{B}} \text{ITE} \quad \frac{\frac{\vdots}{x : T_1 \vdash \text{snd } x : T_3} \text{Snd}}{x : T_1 \vdash \langle \text{if fst } x \text{ then false else true, snd } x \rangle : \mathbb{B} \times T_3} \text{Pair}$$

Now we see that any choice of T_3 will satisfy all of the constraints. So, the valid answers are

$$\begin{array}{l} O : \exists \alpha. \mathbb{B} \times T \vdash \text{unpack } \alpha, x = O \text{ in} \\ \text{pack } \langle \alpha, \langle \text{if fst } x \text{ then false else true, snd } x \rangle \rangle \text{ as } \exists \alpha. \mathbb{B} \times T : \exists \alpha. \mathbb{B} \times T \end{array}$$

for any type T .

Question 3: Abstraction barriers

Determine whether each of the following terms is typeable.

- (a) $\text{unpack } \nu, x = \text{pack } \langle \mathbb{N}, \langle \text{one}, \lambda x : \mathbb{N}. \text{succ } x \rangle \rangle \text{ as } \exists \alpha. \alpha \times (\alpha \rightarrow \alpha) \text{ in } \text{succ } (\text{fst } x)$

Not typeable. We expand the term derivation tree from bottom to top, and then attempt to insert types from the top down as follows.

$$\frac{\frac{\frac{\frac{}{x : \nu \times (\nu \rightarrow \nu) \vdash x : \nu \times (\nu \rightarrow \nu)}}{\text{VAR}}}{\text{Fst}}}{\frac{x : \nu \times (\nu \rightarrow \nu) \vdash \text{fst } x : \nu}{\text{S}}} \quad \text{S} \quad \frac{}{\vdash \text{unpack } \nu, x = \text{pack } \langle \mathbb{N}, \langle \text{one}, \lambda x : \mathbb{N}. \text{succ } x \rangle \rangle \text{ as } \exists \alpha. \alpha \times (\alpha \rightarrow \alpha) \text{ in } \text{succ } (\text{fst } x) : ?} \text{Unpack}$$

However, we cannot use the S rule on $\text{fst } x$ when it has type ν instead of $\mathbb{N}$.

- (b) $\text{unpack } \nu, x = \text{pack } \langle \mathbb{N}, \langle \text{one}, \lambda x : \mathbb{N}. \text{succ } x \rangle \rangle \text{ as } \exists \alpha. \alpha \times (\alpha \rightarrow \alpha) \text{ in } (\text{snd } x) (\text{fst } x)$

Not typeable. We expand the term derivation tree from bottom to top, and then attempt to insert types from the top down as follows.

$$\frac{\frac{\frac{\frac{}{x : \nu \times (\nu \rightarrow \nu) \vdash x : \nu \times (\nu \rightarrow \nu)}}{\text{VAR}}}{\text{Snd}} \quad \frac{\frac{\frac{\frac{}{x : \nu \times (\nu \rightarrow \nu) \vdash x : \nu \times (\nu \rightarrow \nu)}}{\text{VAR}}}{\text{Fst}}}{\frac{x : \nu \times (\nu \rightarrow \nu) \vdash \text{fst } x : \nu}{\text{APP}}} \quad \text{APP} \quad \frac{}{\vdash \text{unpack } \nu, x = \text{pack } \langle \mathbb{N}, \langle \text{one}, \lambda x : \mathbb{N}. \text{succ } x \rangle \rangle \text{ as } \exists \alpha. \alpha \times (\alpha \rightarrow \alpha) \text{ in } (\text{snd } x) (\text{fst } x) : ?} \text{Unpack}$$

However, we cannot use the Unpack rule at the last step, because of the side condition that ν does not appear free in the type.

- (c) $\text{unpack } \nu, x = \text{pack } \langle \mathbb{N}, \langle \text{one}, \lambda x : \mathbb{N}. \text{succ } x \rangle \rangle \text{ as } \exists \alpha. \alpha \times (\alpha \rightarrow \alpha) \text{ in } \text{pack } \langle \nu, \langle (\text{snd } x) (\text{fst } x), \text{snd } x \rangle \rangle \text{ as } \exists \alpha. \alpha \times (\alpha \rightarrow \alpha)$

Typeable. Write

$$P_1 = \text{pack } \langle \mathbb{N}, \langle \text{one}, \lambda x : \mathbb{N}. \text{succ } x \rangle \rangle \text{ as } \exists \alpha. \alpha \times (\alpha \rightarrow \alpha)$$

$$P_2 = \text{pack } \langle \nu, \langle (\text{snd } x) (\text{fst } x), \text{snd } x \rangle \rangle \text{ as } \exists \alpha. \alpha \times (\alpha \rightarrow \alpha)$$

then

$$\frac{\Pi_1 \quad \Pi_2}{\text{unpack } \nu, x = P_1 \text{ in } P_2 : \exists \alpha. \alpha \times (\alpha \rightarrow \alpha)} \text{Unpack}$$

where Π_1 is

$$\frac{\frac{\frac{}{\vdash \text{zero} : \mathbb{N}}{\text{Z}}} \quad \frac{\frac{\frac{}{x : \mathbb{N} \vdash x : \mathbb{N}}{\text{VAR}}}{\text{S}}}{\frac{x : \mathbb{N} \vdash \text{succ } x : \mathbb{N}}{\text{ABS}}} \quad \frac{}{\vdash \text{one} : \mathbb{N}} \quad \frac{}{\vdash \lambda x : \mathbb{N}. \text{succ } x : \mathbb{N} \rightarrow \mathbb{N}} \quad \text{Pair} \quad \frac{}{\vdash \langle \text{one}, \lambda x : \mathbb{N}. \text{succ } x \rangle : \mathbb{N} \times (\mathbb{N} \rightarrow \mathbb{N})} \text{Pack} \quad \frac{}{\vdash P_1 : \exists \alpha. \alpha \times (\alpha \rightarrow \alpha)}$$

and Π_2 is

$$\frac{\frac{\vdots}{x : \nu \times (\nu \rightarrow \nu) \vdash (\mathbf{snd} \, x) (\mathbf{fst} \, x) : \nu} \text{APP} \quad \frac{\vdots}{x : \nu \times (\nu \rightarrow \nu) \vdash \mathbf{snd} \, x : \nu \rightarrow \nu} \text{Snd}}{\frac{x : \nu \times (\nu \rightarrow \nu) \vdash \langle (\mathbf{snd} \, x) (\mathbf{fst} \, x), \mathbf{snd} \, x \rangle : \nu \times (\nu \rightarrow \nu) \text{Pair}}{x : \nu \times (\nu \rightarrow \nu) \vdash P_2 : \exists \alpha. \alpha \times (\alpha \rightarrow \alpha)} \text{Pack}}$$

(some easy parts omitted).

(d) $\lambda P : \exists \alpha. \mathbb{N} \times \alpha \times (\alpha \rightarrow \mathbb{N}) \times (\alpha \rightarrow \mathbb{B})$.

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unpack  $\gamma, x = P$  in
let  $n = \mathbf{fst} (\mathbf{fst} (\mathbf{fst} \, x))$  in
let  $p = \mathbf{snd} (\mathbf{fst} (\mathbf{fst} \, x))$  in
let  $\pi_1 = \mathbf{snd} (\mathbf{fst} \, x)$  in
let  $\pi_2 = \mathbf{snd} \, x$  in
pack  $\langle \gamma, \langle \langle \pi_1 \, p, p \rangle, \pi_1 \rangle, \pi_2 \rangle \rangle$  as  $\exists \alpha. \mathbb{N} \times \alpha \times (\alpha \rightarrow \mathbb{N}) \times (\alpha \rightarrow \mathbb{B})$ 
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Typeable Writing $C = \exists \alpha. \mathbb{N} \times \alpha \times (\alpha \rightarrow \mathbb{N}) \rightarrow (\alpha \rightarrow \mathbb{B})$, the type will be $C \rightarrow C$. Write

$$I[\gamma] = \mathbb{N} \times \gamma \times (\gamma \rightarrow \mathbb{N}) \times (\gamma \rightarrow \mathbb{B}).$$

Then it is easy to prove the following typing judgements

$$\begin{aligned} x : I[\gamma] &\vdash \mathbf{snd} \, x : \gamma \rightarrow \mathbb{B} \\ x : I[\gamma] &\vdash \mathbf{snd} (\mathbf{fst} \, x) : \gamma \rightarrow \mathbb{N} \\ x : I[\gamma] &\vdash \mathbf{snd} (\mathbf{fst} (\mathbf{fst} \, x)) : \gamma \\ x : I[\gamma] &\vdash \mathbf{fst} (\mathbf{fst} (\mathbf{fst} \, x)) : \mathbb{N} \end{aligned}$$

Moreover, writing $\Gamma = x : I[\gamma], \pi_2 : \gamma \rightarrow \mathbb{B}, \pi_1 : \gamma \rightarrow \mathbb{N}, p : \gamma, n : \mathbb{N}$, we can prove

$$\frac{\frac{\frac{\overline{\Gamma \vdash \pi_1 : \gamma \rightarrow \mathbb{N}} \text{VAR} \quad \overline{\Gamma \vdash p : \gamma} \text{VAR}}{\Gamma \vdash \pi_1 \, p : \mathbb{N}} \text{APP} \quad \overline{\Gamma \vdash p : \gamma} \text{VAR}}{\Gamma \vdash \langle \pi_1 \, p, p \rangle : \mathbb{N} \times \gamma} \text{Pair} \quad \frac{\overline{\Gamma \vdash \pi_1 : \gamma \rightarrow \mathbb{N}} \text{VAR}}{\Gamma \vdash \langle \langle \pi_1 \, p, p \rangle, \pi_1 \rangle : \mathbb{N} \times \gamma \times (\gamma \rightarrow \mathbb{N})} \text{Pair} \quad \frac{\Gamma \vdash \langle \langle \pi_1 \, p, p \rangle, \pi_1 \rangle, \pi_2 \rangle : I[\gamma]}{\Gamma \vdash \langle \langle \pi_1 \, p, p \rangle, \pi_1 \rangle, \pi_2 \rangle : I[\gamma]} \text{Pair} \text{VAR}$$

It follows immediately that we have

$$\Gamma \vdash \text{pack } \langle \gamma, \langle \langle \pi_1 \, p, p \rangle, \pi_1 \rangle, \pi_2 \rangle \rangle \text{ as } C$$

Writing M for the entire body of the `unpack` statement, four applications of the `LET` rule give us

$$x : I[\gamma] \vdash M : C$$

whence we can finish the derivation

$$\frac{\frac{\overline{P : C \vdash P : C} \text{VAR} \quad \overline{P : C, x : I[\gamma] \vdash M : C} \text{(above)}}{P : C \vdash \text{unpack } \gamma, x = P \text{ in } M : C} \text{Unpack}}{\vdash \lambda P : C. \text{unpack } \gamma, x = P \text{ in } M : C \rightarrow C} \text{ABS}$$

(e) $\lambda P : \exists \alpha. \alpha \times (\alpha \rightarrow \mathbb{N}) \times (\alpha \rightarrow \mathbb{B})$.

unpack $\gamma, x = P$ in
 let $p = \text{fst}(\text{fst } x)$ in
 let $\pi_1 = \text{snd}(\text{fst } x)$ in
 let $\pi_2 = \text{snd } x$ in $\langle \pi_1 p, \pi_2 p \rangle$

Typeable. Write $I[\gamma] = \gamma \times (\gamma \rightarrow \mathbb{N}) \times (\gamma \rightarrow \mathbb{B})$ and $C = \exists \alpha. I[\alpha]$ as before.

$$\begin{array}{c}
 \frac{\frac{\frac{\Pi_3 \quad P : C, x : I[\gamma], p : \gamma, \pi_1 : \gamma \rightarrow \mathbb{N}, \pi_2 : \gamma \rightarrow \mathbb{B} \vdash \langle \pi_1 p, \pi_2 p \rangle : \mathbb{N} \times \mathbb{B}}{\Pi_2 \quad P : C, x : I[\gamma], p : \gamma, \pi_1 : \gamma \rightarrow \mathbb{N} \vdash \text{let } \pi_2 = \text{snd } x \text{ in } \langle \pi_1 p, \pi_2 p \rangle : \mathbb{N} \times \mathbb{B}} \text{LET}}{\Pi_1 \quad P : C, x : I[\gamma], p : \gamma \vdash \text{let } \pi_1 = \text{snd}(\text{fst } x) \text{ in} \\ \text{let } \pi_2 = \text{snd } x \text{ in } \langle \pi_1 p, \pi_2 p \rangle : \mathbb{N} \times \mathbb{B}} \text{LET} \\
 \frac{\frac{\frac{P : C, x : I[\gamma] \vdash \text{let } p = \text{fst}(\text{fst } x) \text{ in} \\ \text{let } \pi_1 = \text{snd}(\text{fst } x) \text{ in} : \mathbb{N} \times \mathbb{B} \\ \text{let } \pi_2 = \text{snd } x \text{ in } \langle \pi_1 p, \pi_2 p \rangle}{\text{unpack } \gamma, x = P \text{ in} \\ \text{let } p = \text{fst}(\text{fst } x) \text{ in} : \mathbb{N} \times \mathbb{B} \\ \text{let } \pi_1 = \text{snd}(\text{fst } x) \text{ in} : \mathbb{N} \times \mathbb{B} \\ \text{let } \pi_2 = \text{snd } x \text{ in } \langle \pi_1 p, \pi_2 p \rangle} \text{Unpack}}{\lambda P : C. \\ \vdash \text{unpack } \gamma, x = P \text{ in} \\ \text{let } p = \text{fst}(\text{fst } x) \text{ in} : C \rightarrow \mathbb{N} \times \mathbb{B} \\ \text{let } \pi_1 = \text{snd}(\text{fst } x) \text{ in} \\ \text{let } \pi_2 = \text{snd } x \text{ in } \langle \pi_1 p, \pi_2 p \rangle} \text{ABS}
 \end{array}$$

where $\Pi_1, \Pi_2, \Pi_3, \Pi_4, \Pi_5$ are the straightforward proofs of

$$\begin{array}{l}
 P : C, x : I[\gamma] \vdash \text{fst}(\text{fst } x) : \gamma \\
 P : C, x : I[\gamma], p : \gamma \vdash \text{snd}(\text{fst } x) : \gamma \rightarrow \mathbb{N} \\
 P : C, x : I[\gamma], p : \gamma, \pi_1 : \gamma \rightarrow \mathbb{N} \vdash \text{snd } x : \gamma \rightarrow \mathbb{B} \\
 P : C, x : I[\gamma], p : \gamma, \pi_1 : \gamma \rightarrow \mathbb{N}, \pi_2 : \gamma \rightarrow \mathbb{B} \vdash \pi_1 p : \mathbb{N} \\
 P : C, x : I[\gamma], p : \gamma, \pi_1 : \gamma \rightarrow \mathbb{N}, \pi_2 : \gamma \rightarrow \mathbb{B} \vdash \pi_2 p : \mathbb{B}
 \end{array}$$

Question 4: Small-step CBV semantics with existential types

Reduce each of the following terms to a CBV normal form.

(a) $\text{pack } \langle \mathbb{N}, (\lambda x : \mathbb{N}. \text{succ } x) \text{ zero} \rangle \text{ as } \exists \alpha. \alpha$

There is only one small step to do, and it has a single-layer context,

$$\frac{\frac{\overline{(\lambda x : \mathbb{N}. \text{succ } x) \text{ zero} \rightarrow_v \text{succ zero}}^\beta}{\text{pack } \langle \mathbb{N}, (\lambda x : \mathbb{N}. \text{succ } x) \text{ zero} \rangle \text{ as } \exists \alpha. \alpha \rightarrow_v \text{pack } \langle \mathbb{N}, \text{succ zero} \rangle \text{ as } \exists \alpha. \alpha} \text{CTX}_{\text{pack } \langle \mathbb{N}, \bullet \rangle \text{ as } \exists \alpha. \alpha}$$

where $\text{pack } \langle \mathbb{N}, \text{succ zero} \rangle \text{ as } \exists \alpha. \alpha$ is indeed a normal form.

(b) $\text{unpack } \alpha, x = \text{pack } \langle \mathbb{N} \times \mathbb{N}, \langle \text{zero}, \text{one} \rangle \rangle \text{ as } \exists \alpha. \mathbb{N} \text{ in } \text{fst } x$

This term is not typeable, but that is irrelevant for reducing to a normal form. The reduction takes two steps, and needs no context rules:

$$\frac{\text{unpack } \alpha, x = \text{pack } \langle \mathbb{N} \times \mathbb{N}, \langle \text{zero}, \text{one} \rangle \rangle \text{ as } \exists \alpha. \mathbb{N} \text{ in } \text{fst } x \rightarrow_v \text{fst } \langle \text{zero}, \text{one} \rangle}{\text{fst } \langle \text{zero}, \text{one} \rangle \rightarrow_v \text{zero}} \text{UnP}$$

$$\frac{}{\text{fst } \langle \text{zero}, \text{one} \rangle \rightarrow_v \text{zero}} \text{FstP}$$

where zero is indeed a normal form.

(c) $\text{unpack } \theta, x = \text{pack } \langle \mathbb{N}, \langle \text{zero}, \text{one} \rangle \rangle \text{ as } \exists \alpha. \alpha \times \alpha \text{ in}$

let $a_1 = \text{fst } x$ in
 let $b_1 = \text{snd } x$ in
 let $a_2 = \text{succ } (\text{succ } b_1)$ in
 let $b_2 = \text{succ } a_1$ in
 pack $\langle \mathbb{N}, \langle a_2, b_2 \rangle \rangle$ as $\exists \alpha. \alpha \times \alpha$

We'll write the evaluation context at the side just before each reduction step.

unpack $\theta, x = \text{pack } \langle \mathbb{N}, \langle \text{zero}, \text{one} \rangle \rangle$ as $\exists \alpha. \alpha \times \alpha$ in
 let $a_1 = \text{fst } x$ in
 let $b_1 = \text{snd } x$ in
 let $a_2 = \text{succ } (\text{succ } b_1)$ in
 let $b_2 = \text{succ } a_1$ in
 pack $\langle \mathbb{N}, \langle a_2, b_2 \rangle \rangle$ as $\exists \alpha. \alpha \times \alpha$ (•)

$\rightarrow_v$ let $a_1 = \text{fst } \langle \text{zero}, \text{one} \rangle$ in
 let $b_1 = \text{snd } \langle \text{zero}, \text{one} \rangle$ in
 let $a_2 = \text{succ } (\text{succ } b_1)$ in
 let $b_2 = \text{succ } a_1$ in
 pack $\langle \mathbb{N}, \langle a_2, b_2 \rangle \rangle$ as $\exists \alpha. \alpha \times \alpha$ $\left(\begin{array}{l} \text{let } a_1 = \bullet \text{ in} \\ \text{let } b_1 = \text{snd } \langle \text{zero}, \text{one} \rangle \text{ in} \\ \text{let } a_2 = \text{succ } (\text{succ } b_1) \text{ in} \\ \text{let } b_2 = \text{succ } a_1 \text{ in} \\ \text{pack } \langle \mathbb{N}, \langle a_2, b_2 \rangle \rangle \text{ as } \exists \alpha. \alpha \times \alpha \end{array} \right)$

$\rightarrow_v$ let $a_1 = \text{zero}$ in
 let $b_1 = \text{snd } \langle \text{zero}, \text{one} \rangle$ in
 let $a_2 = \text{succ } (\text{succ } b_1)$ in
 let $b_2 = \text{succ } a_1$ in
 pack $\langle \mathbb{N}, \langle a_2, b_2 \rangle \rangle$ as $\exists \alpha. \alpha \times \alpha$ (•)

$\rightarrow_v$ let $b_1 = \text{snd } \langle \text{zero}, \text{one} \rangle$ in
 let $a_2 = \text{succ } (\text{succ } b_1)$ in
 let $b_2 = \text{succ } \text{zero}$ in
 pack $\langle \mathbb{N}, \langle a_2, b_2 \rangle \rangle$ as $\exists \alpha. \alpha \times \alpha$ $\left(\begin{array}{l} \text{let } b_1 = \bullet \text{ in} \\ \text{let } a_2 = \text{succ } (\text{succ } b_1) \text{ in} \\ \text{let } b_2 = \text{succ } a_1 \text{ in} \\ \text{pack } \langle \mathbb{N}, \langle a_2, b_2 \rangle \rangle \text{ as } \exists \alpha. \alpha \times \alpha \end{array} \right)$

$\rightarrow_v$ let $b_1 = \text{one}$ in
 let $a_2 = \text{succ } (\text{succ } b_1)$ in
 let $b_2 = \text{succ } \text{zero}$ in
 pack $\langle \mathbb{N}, \langle a_2, b_2 \rangle \rangle$ as $\exists \alpha. \alpha \times \alpha$ (•)

$\rightarrow_v$ `let  $a_2 = \text{succ}(\text{succ one})$  in`
 `let  $b_2 = \text{succ zero}$  in`
 `pack  $\langle \mathbb{N}, \langle a_2, b_2 \rangle \rangle$  as  $\exists \alpha. \alpha \times \alpha$` ($\bullet$)

$\rightarrow_v$ `let  $b_2 = \text{succ zero}$  in`
 `pack  $\langle \mathbb{N}, \langle \text{succ}(\text{succ one}), b_2 \rangle \rangle$  as  $\exists \alpha. \alpha \times \alpha$` ($\bullet$)

$\rightarrow_v$ `pack  $\langle \mathbb{N}, \langle \text{succ}(\text{succ one}), \text{succ zero} \rangle \rangle$  as  $\exists \alpha. \alpha \times \alpha$`

which is a normal form.